

Harry Nguyen

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EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • **GPA: 3.75/4.0**

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

EXPERIENCE

TiMoto AI | Software Engineer

Sep 2025 – Present

- Led backend, cloud infrastructure, and ML serving for a 3-person engineering team – designed and operated **batch and real-time data pipelines** ingesting and serving production ML workloads at **sub-50ms p99**; **PostgreSQL** + **Redis** data layer with zero data loss at production traffic
- Built **multi-layer observability stack** for production fleet monitoring – **CloudWatch** + **Prometheus** + **Grafana** dashboards covering infra SLOs (uptime, latency) and pipeline correctness metrics; **99.9% uptime SLO**; on-call rotation with runbooks and post-mortems
- Designed and maintained **gRPC inter-service layer**; debugged production deadlock by tracing shared resource acquisition conflicts and redesigning call sequencing – **exactly-once delivery** across distributed services; circuit breaker pattern with auto-rollback
- Architected multi-AZ **ECS Fargate with Terraform IaC** (AWS/GCP multi-cloud) – **44% cost reduction** (\$40–60/month); **vLLM inference engine with PagedAttention** for zero-OOM ML serving at production traffic

Google | Software Engineer Intern – Chrome Browser

May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with **Protocol Buffers** for low-latency inter-process communication – selected Protocol Buffers for schema evolution and cross-language compatibility; shipped to Chrome stable at **sub-50ms p99**, **10K+ req/sec**, **3B+ active users**
- Identified settings navigation as p99 bottleneck at 1,200ms; designed and implemented **lock-free concurrent trie search** eliminating mutex contention – **96% latency reduction**, zero production regressions
- Architected event-driven TypeScript/React system with observer pattern – **68% feature delivery acceleration** across **25K+ lines** of Chromium at **95% test coverage**; shipped to **3B+ active users**
- Delivered design documents and code reviews at **95% test coverage**; changes adopted into production branch reviewed by senior Chrome engineers

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless BaaS on AWS – selected JWT over session auth for horizontal scalability; auto-scaling policy supporting **500+ concurrent users**, **90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+ on large datasets; redesigned with **PostgreSQL composite indexing** achieving **sub-100ms** for 10,000+ records

PROJECTS

Pulumi – Contributor | *Go, TypeScript, Multi-cloud IaC (AWS/Azure/GCP)*

github.com/pulumi

- Submitted **Go CLI features** enabling **multi-cloud (AWS/Azure/GCP) provisioning** – developer tooling contribution under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer – studied linearizability guarantees and correctness under concurrent operations and partial failures

SKILLS

Languages: C++, Rust, TypeScript, Python, Go, Java, JavaScript, SQL, Bash

Distributed Systems: gRPC, Protocol Buffers, Exactly-once semantics, Circuit breakers, Fault tolerance, Multi-AZ, Event-driven architecture, Kafka

Cloud & Infrastructure: AWS (ECS Fargate, EKS, EC2, S3, RDS), GCP, Terraform, **Kubernetes**, Docker, CI/CD, CloudWatch, Prometheus, Grafana

Frameworks & Databases: FastAPI, Django, Node.js, React, REST APIs, PostgreSQL, Redis, MongoDB

ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LLM APIs (Claude, GPT), LangChain, LLM-as-a-judge evaluation, PyTorch

AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor